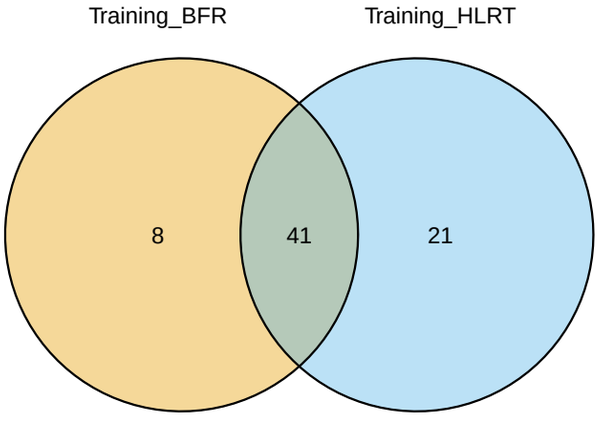
proton transmembrane transporter activity

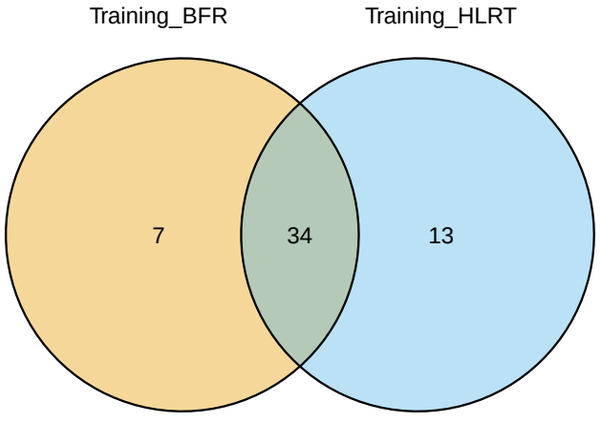
Training_BFR: NES 2.52, padj 2.6e−08
Training_HLRT: NES 2.64, padj 3.1e−10



Shared: ATP5F1A, ATP5F1C, ATP5F1D, ATP5MG, ATP5PB, ATP5PD, ATP5PO, COX4I1, COX5B, COX6B1, COX7A1, COX7A2, COX7B, CYC1, GHITM, LETM1, MT−CO3, NDUFA4, NDUFA9, NDUFB10, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, SLC16A1, SLC16A3, SLC25A12, SLC25A3, S...

oxidoreduction−driven active transmembrane transporter activity

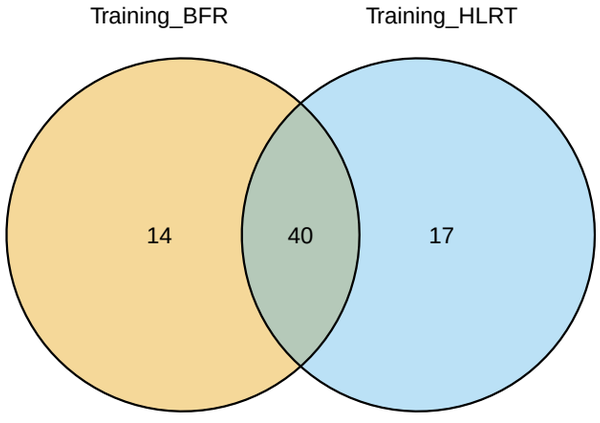
Training_BFR: NES 2.55, padj 1.9e−07
Training_HLRT: NES 2.85, padj 2.9e−12



Shared: COX4I1, COX5B, COX6B1, COX7A1, COX7A2, COX7A2L, COX7B, CYBRD1, CYC1, MT−CO3, NDUFA10, NDUFA12, NDUFA4, NDUFA9, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, UQCRC1, UQCRFS1, UQCRH

electron transfer activity

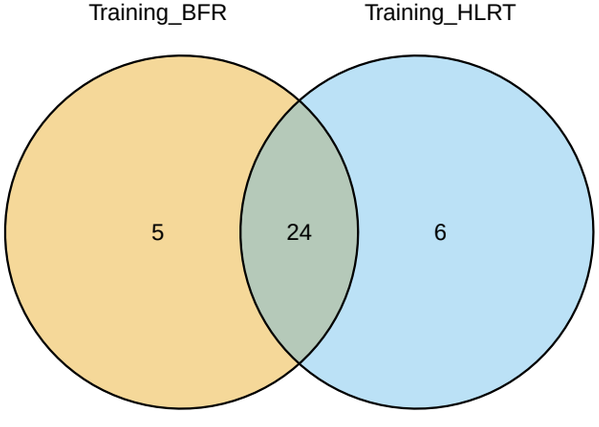
Training_BFR: NES 2.38, padj 1.7e−06
Training_HLRT: NES 2.46, padj 3.2e−08



Shared: AKR7A2, CIAPIN1, COX4I1, COX5B, COX6B1, COX7A1, COX7A2, COX7A2L, COX7B, CYBB, CYC1, ETFA, ETFB, IDH3B, ME2, MT−CO3, NDUFA12, NDUFA4, NDUFA9, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS8, NDUFV1, NDUFV2, SDHA, UQCR11, UQCRC1, UQCRF...

NADH dehydrogenase (ubiquinone) activity

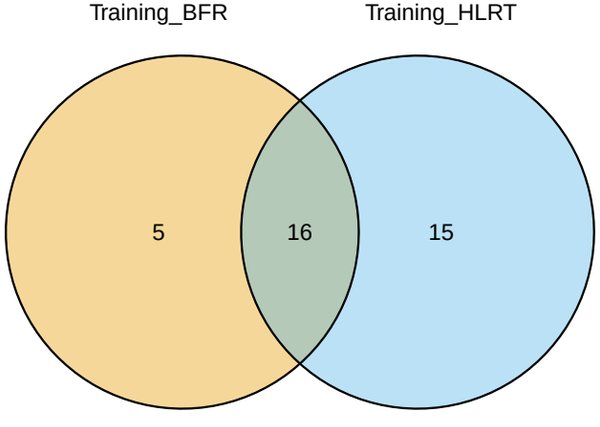
Training_BFR: NES 2.26, padj 0.00017
Training_HLRT: NES 2.65, padj 9.5e−09



Shared: MT−ND1, NDUFA10, NDUFA12, NDUFA4, NDUFA8, NDUFA9, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2

catalytic activity, acting on a tRNA

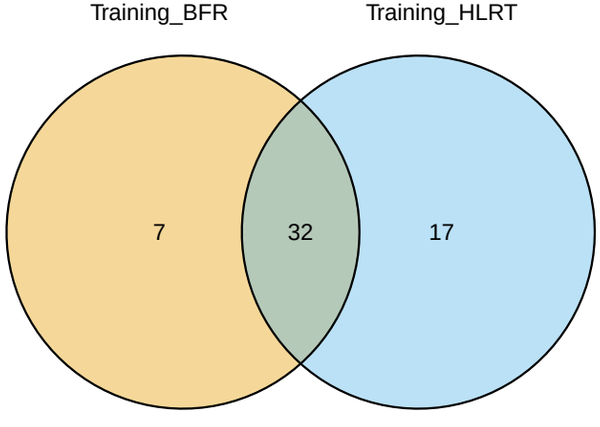
Training_BFR: NES 2.27, padj 0.00017
Training_HLRT: NES 2.17, padj 0.00055



Shared: GGT5, HARS1, HARS2, IARS1, IARS2, LARS1, LARS2, NARS1, NSUN2, QARS1, SARS1, SARS2, TARS1, TARS3, TRMT10C, TRPT1

ligase activity

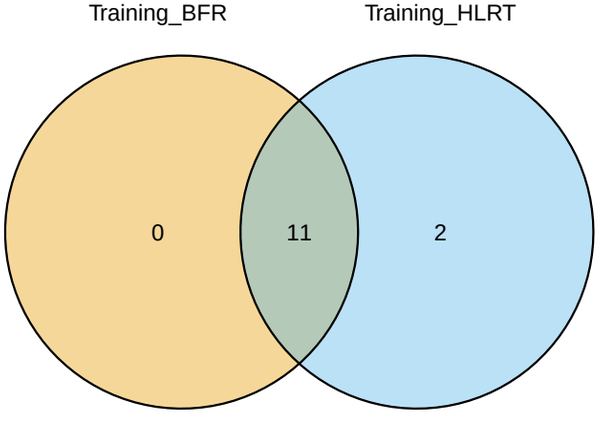
Training_BFR: NES 2.41, padj 1.9e−07
Training_HLRT: NES 1.95, padj 0.00056



Shared: ACSF3, ACSL1, ACSS1, AK3, ATP5F1A, ATP5F1B, ATP5F1C, ATP5F1D, ATP5ME, ATP5MG, ATP5PB, ATP5PD, ATP5PO, CARNS1, HARS1, HARS2, IARS1, IARS2, LARS1, LARS2, MCCC2, NARS1, PCCA, PCCB, QARS1, SARS1, SARS2, SUCLA2, SUCLG1, SUCLG2, TARS1, TARS3

calcium ion transmembrane transporter activity

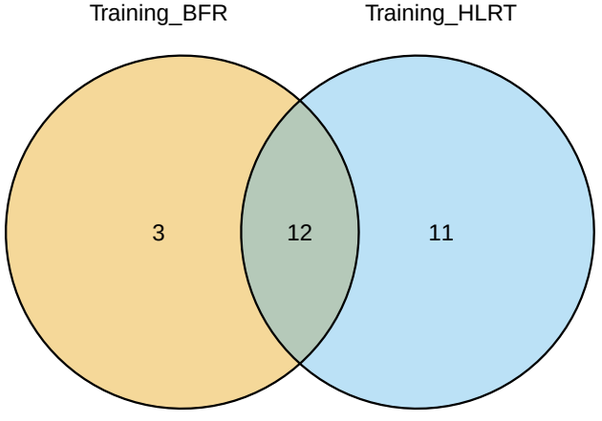
Training_BFR: NES −2.36, padj 9.9e−05
Training_HLRT: NES −2.19, padj 0.00094



Shared: ATP2A1, ATP2A2, ATP2B2, ATP2C1, CACNA1S, CACNA2D1, CACNB1, CACNG1, CUL5, RYR1, TMCO1

aminoacyl−tRNA ligase activity

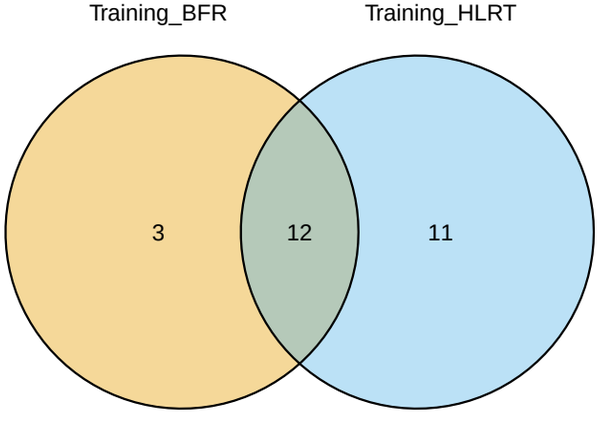
Training_BFR: NES 2.25, padj 0.0003
Training_HLRT: NES 2.11, padj 0.0013



Shared: HARS1, HARS2, IARS1, IARS2, LARS1, LARS2, NARS1, QARS1, SARS1, SARS2, TARS1, TARS3

ligase activity, forming carbon−oxygen bonds

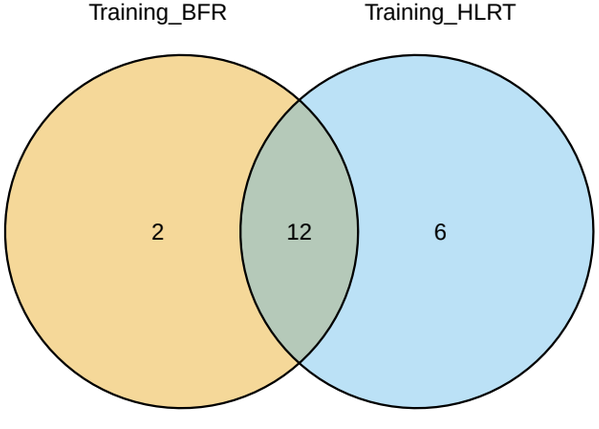
Training_BFR: NES 2.25, padj 0.0003
Training_HLRT: NES 2.11, padj 0.0013



Shared: HARS1, HARS2, IARS1, IARS2, LARS1, LARS2, NARS1, QARS1, SARS1, SARS2, TARS1, TARS3

oxidoreductase activity, acting on NAD(P)H

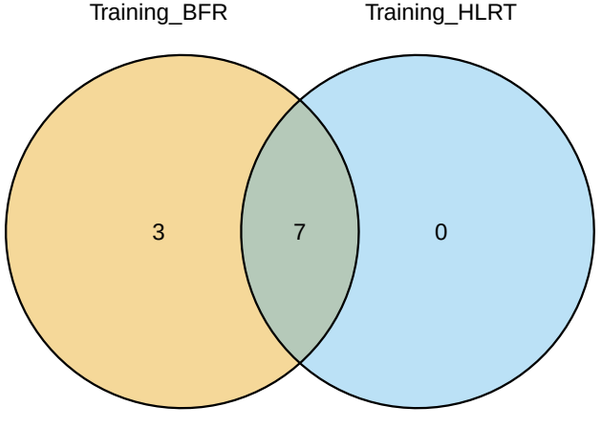
Training_BFR: NES 2.30, padj 0.0003
Training_HLRT: NES 2.06, padj 0.002



Shared: AIFM1, AKR1C3, CBR1, CBR4, CYBB, NDUFA9, NDUFS1, NDUFS3, NDUFS8, NDUFV2, NNT, RTN4IP1

calcium channel regulator activity

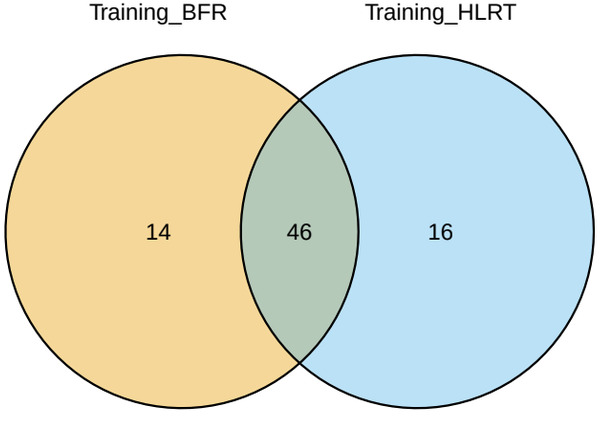
Training_BFR: NES −2.10, padj 0.0062
Training_HLRT: NES −2.03, padj 0.0052



Shared: ATP2A2, CACNB1, CACNG1, GSTM2, PHPT1, STIM1, YWHAE

active transmembrane transporter activity

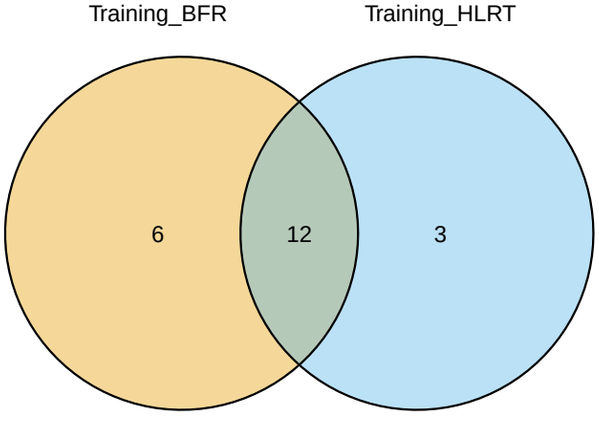
Training_BFR: NES 1.74, padj 0.012
Training_HLRT: NES 2.19, padj 1.5e−06



Shared: ABCB7, ATP1A1, COX4I1, COX5B, COX6B1, COX7A1, COX7A2, COX7A2L, COX7B, CYBRD1, CYC1, GHITM, LETM1, MT−CO3, NDUFA10, NDUFA12, NDUFA4, NDUFA9, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, SLC16A1, SLC16A3, SLC25...

metal ion transmembrane transporter activity

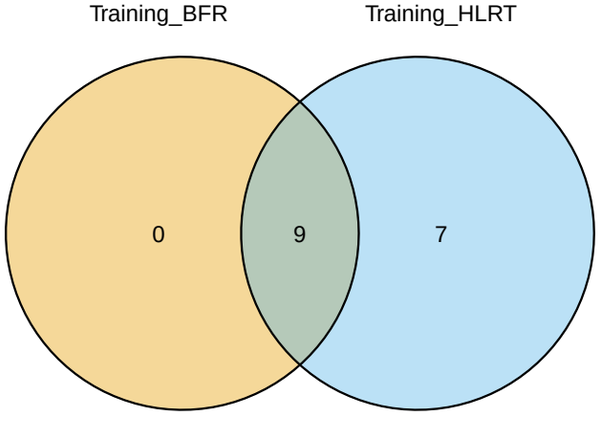
Training_BFR: NES −2.31, padj 9.9e−05
Training_HLRT: NES −1.84, padj 0.017



Shared: ATP2A1, ATP2A2, ATP2B2, ATP2C1, CACNA1S, CACNA2D1, CACNB1, CACNG1, CUL5, RYR1, RYR2, TMCO1

NAD binding

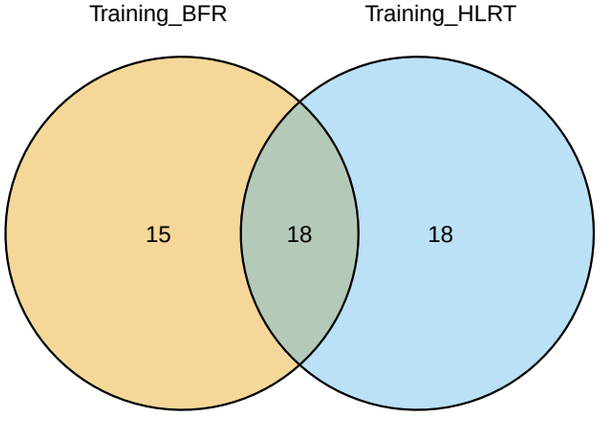
Training_BFR: NES 1.91, padj 0.026
Training_HLRT: NES 2.06, padj 0.0018



Shared: ALDH1B1, HSD17B8, IDH3A, IDH3B, IDH3G, ME2, NNT, SIRT3, UGDH

transcription regulator activity

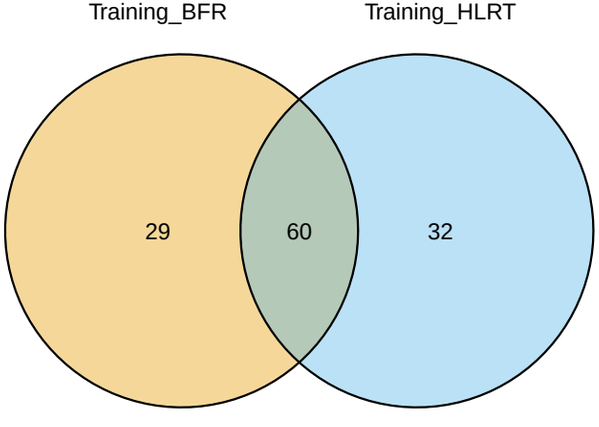
Training_BFR: NES −1.63, padj 0.026
Training_HLRT: NES −1.80, padj 0.0031



Shared: ACSS2, COPS2, CRYM, EDF1, ENO1, FHL3, MECP2, MLIP, PARK7, PBXIP1, PFDN5, PRKN, PRMT5, PURA, STAT5B, TSC22D1, UBE3A, WDR77

adenyl nucleotide binding

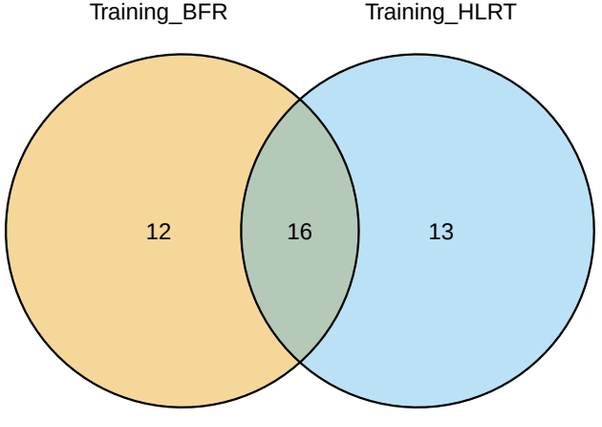
Training_BFR: NES 1.40, padj 0.026
Training_HLRT: NES 1.37, padj 0.028



Shared: ABCB7, ACSL1, ACTR3, AFG3L2, AK2, AK3, AKT2, ALDH1B1, ATP1A1, ATP5F1A, ATP5F1D, BCKDK, CAPRIN1, CARNS1, CBR4, CKMT2, CLPB, DECR1, DNAJA4, EHD1, GLUD1, HK1, HSD17B8, HSPA9, HSPD1, HSPE1, IARS1, IARS2, IDH3A, IDH3B, IDH3G, LARS1, LONP1, ME2, MYH11, MYH9, NADK2, NDUFV1, NME1, NNT, PCCA, PCCB, PDK2, ...

transcription coregulator activity

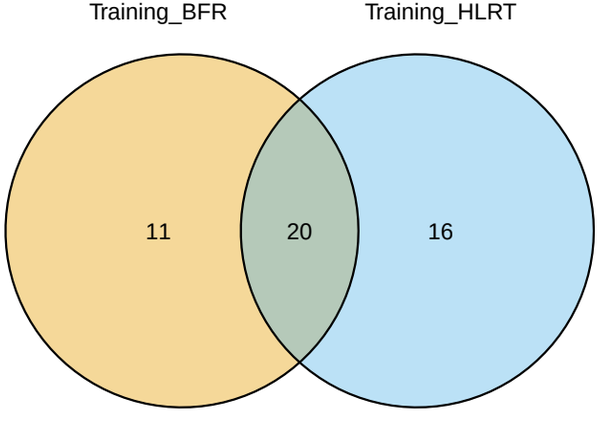
Training_BFR: NES −1.68, padj 0.029
Training_HLRT: NES −1.77, padj 0.008



Shared: ACSS2, COPS2, CRYM, EDF1, ENO1, FHL3, MECP2, MLIP, PARK7, PBXIP1, PFDN5, PRKN, PRMT5, TSC22D1, UBE3A, WDR77

ubiquitin−protein transferase activity

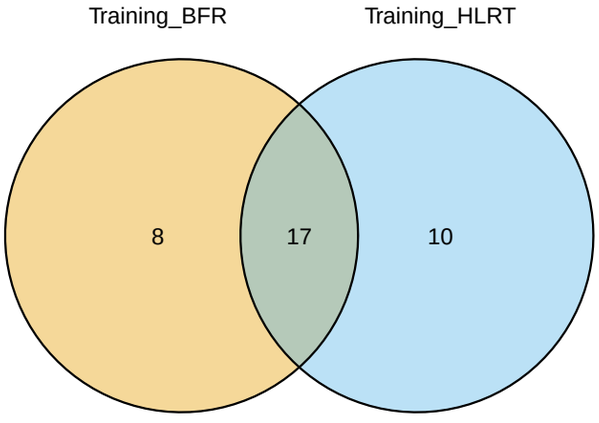
Training_BFR: NES −1.71, padj 0.036
Training_HLRT: NES −1.81, padj 0.006



Shared: ARIH1, ARIH2, CDC42, CUL3, CUL5, HECTD1, HUWE1, MKRN2, PRKN, RNF114, RNF126, RNF170, RNF181, STUB1, TRIM72, UBE2B, UBE2K, UBE2R2, UBE3A, UBR4

ubiquitin protein ligase activity

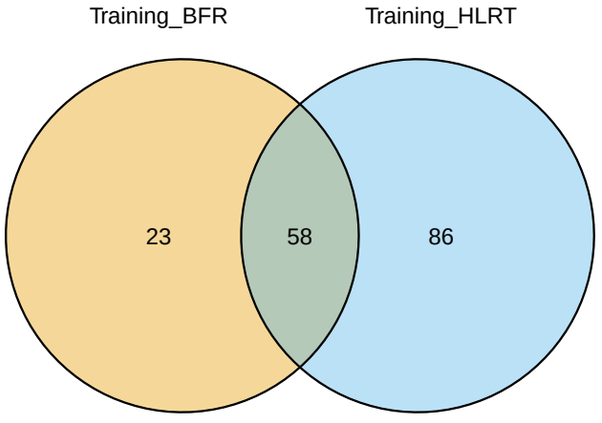
Training_BFR: NES −1.77, padj 0.038
Training_HLRT: NES −1.71, padj 0.028



Shared: ARIH1, ARIH2, CDC42, CUL3, HECTD1, HUWE1, MKRN2, PRKN, RNF114, RNF126, RNF170, RNF181, STUB1, TRIM72, UBE2K, UBE3A, UBR4

transferase activity

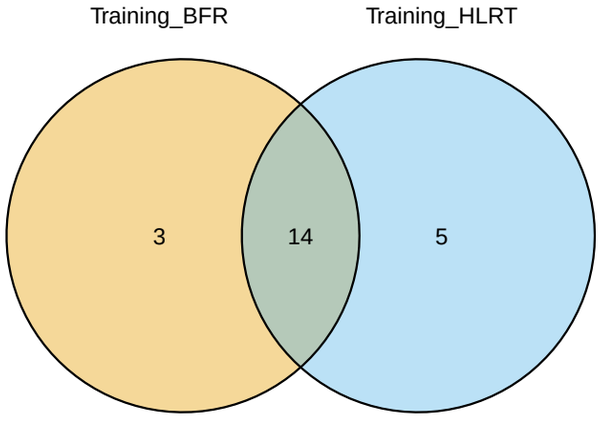
Training_BFR: NES −1.34, padj 0.039
Training_HLRT: NES −1.33, padj 0.025



Shared: AGPS, ARIH1, ARIH2, ART3, CAMK2A, CD38, CERT1, CUL5, DGKZ, FGGY, GGPS1, GMPPA, GNPAT, GPHN, GPT, GPT2, GSTM1, GSTM2, GSTM3, HAS1, HECTD1, KYAT1, MAP2K1, MAP3K20, MAPK12, MAPK14, METTL5, MOCS3, MYLK2, MYLK3, NEK7, PCMT1, PCYT2, PFKM, PIK3R1, POLR2H, PRDX6, PRKAG3, PRKN, PRMT5, PTGES3, RBKS, RNF170...

iron−sulfur cluster binding

Training_BFR: NES 1.76, padj 0.04
Training_HLRT: NES 2.02, padj 0.0024



Shared: ACO2, BOLA2, CIAPIN1, CISD1, CISD3, FDX1, GLRX5, ISCU, NDUFS1, NDUFS8, NDUFV1, NDUFV2, NUBP1, UQCRFS1